

Lecture 4

Microeconomic Theory 3 – ECON 709
Fall 2026

based on: Rubinstein, lecture 2, lecture 4 (proof of Debreu's Theorem); MWG,
sections 1.B, 3.C

Recap' Lecture 3

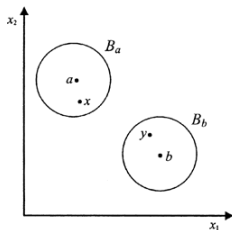
- ▶ A rational preference relation has a utility representation, if X is countable.
- ▶ What are the conditions on a rational preference relation to have a utility representation when X is uncountable?
 - ▶ We saw an example of lexicographic preferences over $X = [0, 1] \times [0, 1]$ which do not have a utility representation.
 - ▶ We are going to show that "continuity" of a rational preference relation is sufficient for a utility representation to exist.

Continuity of $\succsim$: Definition C1

- ▶ Intuition: if $a \in X$ is strictly preferred over $b \in X$, then small deviations from a are also strictly preferred over (small deviations from) b ; small deviations do not change ordering
- ▶ a ball around $a \in X$ with radius $r > 0$, denoted as $Ball(a, r)$, is the set of all points in X that are distanced less than r from a

Definition 3 (C1)

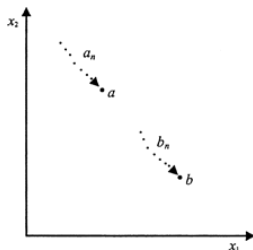
A rational preference relation $\succsim$ on X is continuous if whenever $a \succ b$ (namely, it is not true that $b \succsim a$), there are balls B_a and B_b around a and b , respectively, such that for all $x \in B_a$ and $y \in B_b$, $x \succ y$.



Continuity of $\succsim$: Definition C2

Definition 4 (C2)

A rational preference relation $\succsim$ on X is continuous if it is preserved under limits. That is, for any sequence of pairs $\{(a_n, b_n)\}_{n=1}^{\infty}$ with $a_n \succsim b_n$ for all n , $a_n \rightarrow a$ and $b_n \rightarrow b$, we have $a \succsim b$.



- C2 says the preferences cannot exhibit "jumps", e.g. if you prefer each element in $\{a_n\}_{n=1}^{\infty}$ to the corresponding element in $\{b_n\}_{n=1}^{\infty}$ (because of the convergence, the sequence gets arbitrarily close to a and b), you cannot reverse the preferences in the limit.

Equivalence of C1 and C2

Proposition 6

A rational preference relation on X satisfies C1 if and only if it satisfies C2.

Proof: C1 $\Rightarrow$ C2.

- ▶ Assume that $\succsim$ on X is continuous according to C1.
- ▶ Let $\{(a_n, b_n)\}_{n=1}^{\infty}$ be a sequence of pairs of elements in X with $a_n \succsim b_n$ for all n , $a_n \rightarrow a$ and $b_n \rightarrow b$.
- ▶ Suppose it is not true that $a \succsim b$ and thus $b \succ a$.
- ▶ Then, by C1, there exist two balls B_a and B_b around a and b , respectively, such that for all $y \in B_b$ and $x \in B_a$, $y \succ x$.
- ▶ Since $a_n \rightarrow a$ and $b_n \rightarrow b$, there is an integer N large enough such that for all $n > N$, both $b_n \in B_b$ and $a_n \in B_a$.
- ▶ Therefore, for all $n > N$, we have $b_n \succ a_n$, a contradiction.



Proof: $C2 \Rightarrow C1$.

- ▶ Assume that $\succsim$ is continuous according to $C2$.
- ▶ Let $a \succ b$.
- ▶ Assume by contradiction that $C1$ is not satisfied, that is for any balls B_a and B_b , there exist $x \in B_a$ and $y \in B_b$ such that $y \succsim x$.
- ▶ Then, for all n , there exist $a_n \in Ball(a, 1/n)$ and $b_n \in Ball(b, 1/n)$ such that $b_n \succsim a_n$.
- ▶ The sequence $\{(b_n, a_n)\}_{n=1}^{\infty}$ converges to (b, a) .
- ▶ Then, by $C2$, $b \succsim a$, a contradiction and $C1$ must be satisfied.



Continuity of $\succsim$: Definition C3

Definition 5 (C3)

A rational preference relation $\succsim$ is continuous if, for all $x \in X$, the upper contour set $\{y \in X : y \succsim x\}$ and the lower contour set $\{y \in X : x \succsim y\}$ are closed.

- ▶ reminder: a set is closed if it includes its boundaries
- ▶ Given a preference relation $\succsim$, the upper contour set of $x \in X$ contains all $y \in X$ which are weakly preferred to x , and the lower contour set of x contains all $y \in X$ to which x is weakly preferred.
- ▶ Given a rational preference relation, the union of the lower and upper contour set of any x is equal to X (because of completeness and reflexivity), and the intersection is the indifference set with x .

Equivalence of C1 and C3

Proposition 7

A rational preference relation $\succsim$ on X satisfies C3 if and only if it satisfies C1.

Proof.

- ▶ First note that for all $x \in X$,
 $X \setminus \{y \in X : x \succsim y\} = \{y \in X : y \succ x\}$ and
 $X \setminus \{y \in X : y \succsim x\} = \{y \in X : x \succ y\}$
(complements to $\{y \in X : x \succsim y\}$ and $\{y \in X : y \succsim x\}$,
respectively)
by completeness and reflexivity of $\succsim$.
- ▶ $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open sets if and only if $\{y \in X : x \succsim y\}$ and $\{y \in X : y \succsim x\}$ are closed sets.

Proof cont'd

- ▶ Show:
 $\succsim$ satisfies C1 $\Rightarrow \{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open sets for all $x \in X$ ($\Rightarrow \succsim$ satisfies C3):
 - ▶ Let $a \in \{y \in X : y \succ x\}$.
 - ▶ By C1, there exists $Ball(a, r)$ such that $Ball(a, r) \subseteq \{y \in X : y \succ x\}$.
 - ▶ This is the definition for $\{y \in X : y \succ x\}$ to be an open set.
 - ▶ Equivalently, for $\{y \in X : x \succ y\}$.

Proof cont'd

- ▶ ($\succsim$ satisfies C3 $\Rightarrow$) $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open sets for all $x \in X \Rightarrow \succsim$ satisfies C1:
 - ▶ We use the notation $B \succsim x$ to mean that $y \succsim x$ for all $y \in B$.
 - ▶ Consider $a, b \in X$ such that $a \succ b$.
 - ▶ Assume there exists $z \in X$ such that $a \succ z \succ b$.
 - ▶ Then $a \in \{x \in X : x \succ z\}$ and $b \in \{x \in X : z \succ x\}$.
 - ▶ $\{x \in X : x \succ z\}$ and $\{x \in X : z \succ x\}$ are open sets.
 - ▶ Thus there exists $\epsilon > 0$ such that $Ball(a, \epsilon) \subset \{x \in X : x \succ z\}$ and $Ball(b, \epsilon) \subset \{x \in X : z \succ x\}$, that is $Ball(a, \epsilon) \succ z$ and $z \succ Ball(b, \epsilon)$.
 - ▶ By transitivity, $x \succ y$ for all $x \in Ball(a, \epsilon)$ and all $y \in Ball(b, \epsilon)$.

- ▶ Assume there does not exist $z \in X$ such that $a \succ z \succ b$.
- ▶ By assumption $b \in \{x \in X : a \succ x\}$ and $a \in \{x \in X : x \succ b\}$.
- ▶ $\{x \in X : a \succ x\}$ and $\{x \in X : x \succ b\}$ are open sets.
- ▶ Thus there exists $\epsilon > 0$ such that $Ball(b, \epsilon) \subset \{x \in X : a \succ x\}$ and $Ball(a, \epsilon) \subset \{x \in X : x \succ b\}$, that is $a \succ Ball(b, \epsilon)$ and $Ball(a, \epsilon) \succ b$.
- ▶ Since there is no $z \in X$ (and thus not in $Ball(b, \epsilon)$ and not in $Ball(a, \epsilon)$) such that $a \succ z \succ b$, it must be that $Ball(a, \epsilon) \precsim a$ and $b \precsim Ball(b, \epsilon)$.
- ▶ Thus $Ball(a, \epsilon) \precsim a \succ b \precsim Ball(b, \epsilon)$.
- ▶ And by transitivity, $x \succ y$ for all $x \in Ball(a, \epsilon)$ and all $y \in Ball(b, \epsilon)$.



Our example of lexicographic preferences was not continuous!

- ▶ Remember the example:
Let $X = [0, 1] \times [0, 1]$ and $\succsim_L$ be defined as $(x_1, x_2) \succsim_L (y_1, y_2)$ iff $x_1 > y_1$ or both $x_1 = y_1$ and $x_2 \geq y_2$.
- ▶ Then $a = (0, 1) \succ (0, 0) = b$ but there are no B_a and B_b such that $x \succ y$ for all $x \in B_a$ and $y \in B_b$ because e.g. $(y_1, 0) \succ a$ for $y_1 > 0$.
- ▶ Thus C1 does not hold and $\succsim_L$ is not continuous.

Debreu's Theorem (1954)

Debreu shows that continuity of a rational preference relation is sufficient for the existence of a utility representation.

Proposition 8

Assume X is a convex subset of $\mathbb{R}^n$. Let $\succsim$ be a continuous and rational preference relation on X . Then there is a continuous utility function $u(x)$ which represents $\succsim$.

- ▶ classical result in economic theory
- ▶ we will prove the result with the following additional simplifying assumptions:
 - ▶ $X = \mathbb{R}_+^K = \{x = (x_1, \dots, x_K) \mid \text{for all } k, x_k \geq 0\}$
 - ▶ $\succsim$ is monotonic, i.e. for all x and all $y \in X$, if $x_k \geq y_k$ for all k , then $x \succsim y$, and if $x_k > y_k$ for all k , then $x \succ y$.
"More is better", a common assumption in economics, discussed in more detail later
- ▶ for a proof in the more general setting, see Debreu (1954, 1960)

Proof of Proposition 8 with $X = \mathbb{R}_+^K$ and monotonicity

Proof strategy:

- ▶ construct one function that is continuous and that represents the given preferences
- ▶ first construct a function $u : X \rightarrow \mathbb{R}$, second show that u represents the preferences, third show that u is continuous
- ▶ How do we construct the utility representation here?
For every $x \in X$, we will find one unique indifferent bundle on the main diagonal.
Then we assign to each bundle the amount in its indifferent bundle on the main diagonal as utility value.
(That is, we map all bundles in an order-preserving way onto the main diagonal line and assign each bundle its main diagonal value as utility value.)

First, construct function $u : X \rightarrow \mathbb{R}$.

1. Show that for every $x \in X$, there is $m^* \in X$ on the main diagonal (equal quantities of all commodities), s.t. $x \sim m^*$ (mapping all bundles onto the diagonal):
 - ▶ by monotonicity, $x \succsim 0 = (0, \dots, 0)$ and $M = (\max_k \{x_k\}, \dots, \max_k \{x_k\}) \succsim x$
 - ▶ both 0 and M are on the main diagonal

- construct inductively the sequence $\{(y^n, z^n, m^n)\}$ of elements in X as follows:

Start with $y^0 = M$, $z^0 = 0$ and the midpoint $m^0 = 1/2y^0 + 1/2z^0$.

If $m^n \sim x$, then we found m^* .

If $m^n \succ x$, let $y^{n+1} = m^n$ and $z^{n+1} = z^n$.

If $x \succ m^n$, let $y^{n+1} = y^n$ and $z^{n+1} = m^n$.

(One of the above must be true because of completeness.)

In any case define $m^{n+1} = 1/2y^{n+1} + 1/2z^{n+1}$.

If none of the points $m^n \sim x$, then $y^n \succ x \succ z^n$ for all n .

Both sequences (y^n) and (z^n) converge to some m^* (the same!) on the main diagonal between 0 and M

because $z^n \leq z^{n+1}$ and $y^{n+1} \leq y^n$ with an inequality for one of them, and $z^{n+1} < y^{n+1}$ for all n

(the distance between y^n and z^n is converging to zero).

Since $\succsim$ is continuous, we know that $\{(y^n, x^n)\}$ with $x^n = x$ and $y^n \succsim x$ for all n , $y^n \rightarrow m^*$ implies $m^* \succsim x$ and $\{(x^n, z^n)\}$ with $x^n = x$ and $x \succsim z^n$ for all n , $z^n \rightarrow m^*$ implies $x \succsim m^*$.

Thus $m^* \sim x$.

2. For every x there is a unique m^* on the main diagonal such that $m^* \sim x$ (we check that we mapped each bundle to exactly one point on the main diagonal):
 - Let $m^* \sim x$. For every $m \ll m^*$ (m has every element strictly smaller than m^*) on the main diagonal, $m^* \succ m$ by monotonicity and hence $x \succ m$ by transitivity. Similarly for $m \gg m^*$.
3. Denote the unique m^* on the diagonal such that $m^* \sim x$ by $(t(x), \dots, t(x))$. Let $u(x) = t(x)$. (assigning the utility value to x)

Second, verify that the function $u : X \rightarrow \mathbb{R}$ is a utility representation of $\succsim$.

- ▶ $x \succsim y$
 $\Leftrightarrow (t(x), \dots, t(x)) \succsim (t(y), \dots, t(y))$ (by transitivity)
 $\Leftrightarrow t(x) \geq t(y)$ (by monotonicity)
 $\Leftrightarrow u(x) \geq u(y)$

Third, prove that u is continuous.

- ▶ Reminder: A function $u : X \rightarrow \mathbb{R}$ is continuous if for all x and any sequence (x^n) with $x^n \rightarrow x$, we have $u(x^n) \rightarrow u(x)$.
- ▶ Let (x^n) be a sequence converging to x and assume by contradiction that $t(x^n)$ does not converge to $t(x)$.
- ▶ Since $x^n \rightarrow x$, there is a number M^* s.t. $(M^*, \dots, M^*) \succ x^n$ for all n .
- ▶ Then all members of the sequence $(t(x^n))$ are in the compact interval $[0, M^*]$, as $(t(x^n), \dots, t(x^n)) \sim x^n$.

- ▶ As $t(x^n) \not\rightarrow t(x)$ there is an $\epsilon > 0$ s.t. for every integer N there is $n > N$ such that $t(x^n)$ is outside the interval $(t(x) - \epsilon, t(x) + \epsilon)$ and thus there must be infinitely many t -values outside of that interval.
- ▶ Then, there is an infinite subsequence of (x^n) with all t -values in $[t(x) + \epsilon, M^*]$ (or all t -values in $[0, t(x) - \epsilon]$).
- ▶ This subsequence has an infinite subsequence (y^n) with $(t(y^n))$ converging to some $t' > t(x)$. (by Bolzano-Weierstrass-Theorem)
- ▶ Now, $y^n \rightarrow x$ (as (y^n) is a subsequence of (x^n) which converges to x) and $(t(y^n), \dots, t(y^n)) \rightarrow (t', \dots, t')$.
- ▶ Since $y^n \sim (t(y^n), \dots, t(y^n))$ for all n , $x \sim (t', \dots, t')$ by the continuity of preferences.
- ▶ However, $x \sim (t(x), \dots, t(x))$ and $t' > t(x)$, a contradiction. □

Any continuous utility function represents a continuous and rational $\succsim$

Proposition 9

If $\succsim$ on X is represented by a continuous utility function u , then $\succsim$ is rational and continuous.

Proof.

- ▶ $\succsim$ is rational as rationality is a necessary condition for a utility representation to exist
- ▶ $\succsim$ is also continuous (we show that C1 is satisfied):
 - ▶ Consider $a \succ b$. Then $u(a) > u(b)$.
 - ▶ Let $\epsilon = (u(a) - u(b))/2$. By the continuity of u , there is a $\delta > 0$ such that for all x distanced less than δ from a , $u(x) > u(a) - \epsilon$, and for all y distanced less than δ from b , $u(y) < u(b) + \epsilon$.
 - ▶ Thus, for all x and y within the balls of radius δ around a and b , respectively, $x \succ y$.

